

running stuff on rc

a) using `sinteractive` on our partition

```
# start a job with 20 cpu cores, 4g ram-per-cpu core, and one "full" h100 gpu:
sinteractive --partition=blanca-blast-lecs --account=blanca-blast-lecs --qos=blanca-blast-lecs --
--nodes=1 --ntasks=20 --mem-per-cpu=4000m --gres=gpu:h100_7g.80gb
```

maria said that we should use 5 `ntasks` per 20 gb of gpu usage. therefore, the complete one (80 gb) should have 20 workers. however, rohan might mention u in the slack channel if u do this lol.

alternative (in case that the 80 gb is being used by someone else) - using `sinteractive` on the `clearlab` partition

both partitions work for the transcription step. the `clearlab` one is not useful for the diarization pipeline since it needs at least 80 gb of memory, which none of those gpus have. what i did, for speeding things up, was using all of these gpus - plus the ones in our partition.

```
sinteractive --partition=blanca-clearlab1 --qos=blanca-clearlab1 --account=blanca-clearlab1 --
nodelist=bgpu-g4-u30 --gpu=1
```

b) add the gpu to the environment

```
export CUDA_VISIBLE_DEVICES=""
```

c) load the modules needed

```
module load cuda/12.1.1.lua
```

```
module load ffmpeg/4.4
```

```
module load uv
```

d) add the dependencies needed for correct gpu detection/usage

```
uv sync --extra gpu
```

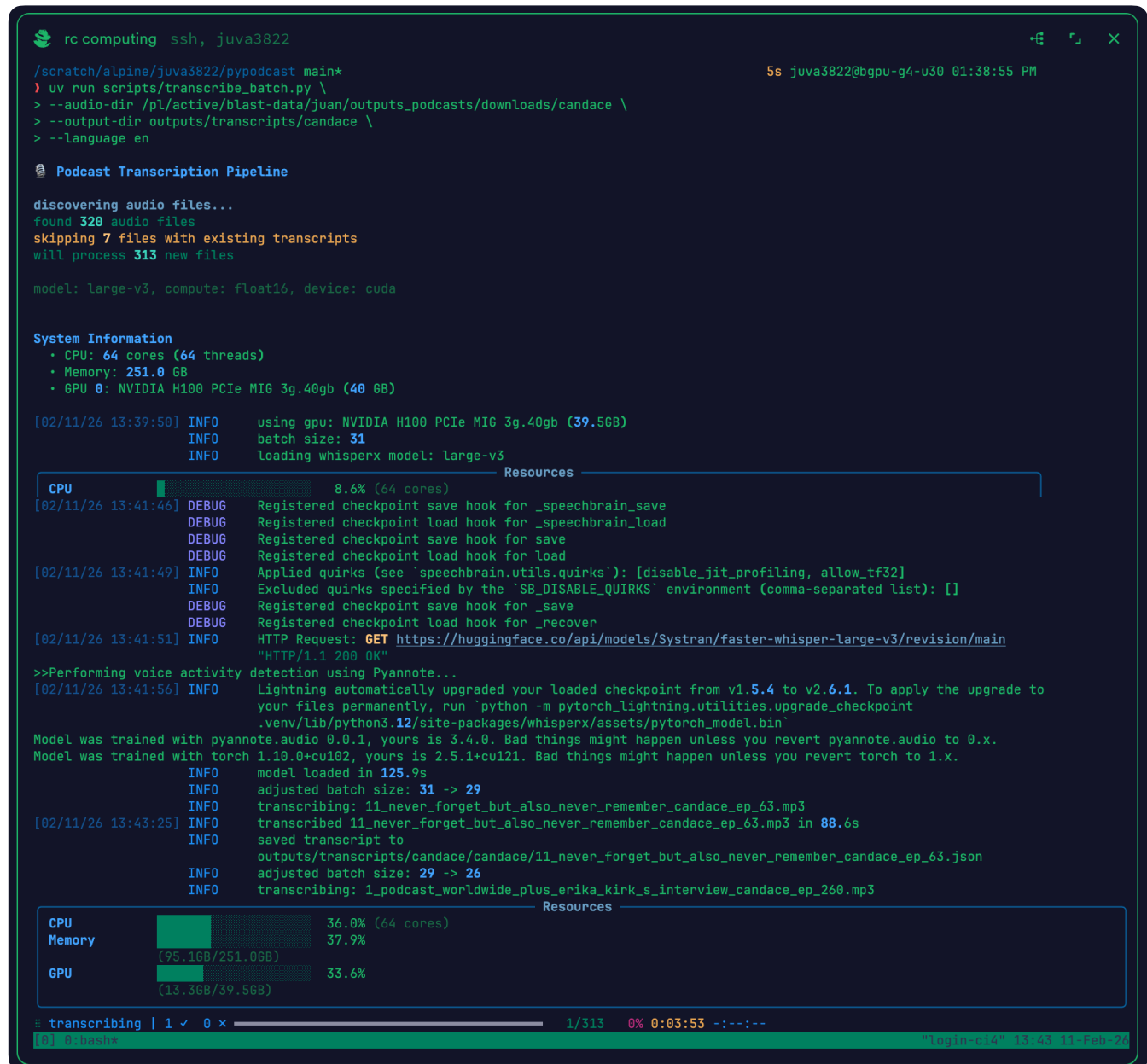
e) transcription

```
wget -O
/scratch/alpine/juva3822/.torchcache/hub/checkpoints/wav2vec2_fairseq_base_ls960_asr_ls960.pth
https://download.pytorch.org/torchaudio/models/wav2vec2_fairseq_base_ls960_asr_ls960.pth
```

the path after `--audio-dir` is where the audio files i downloaded are stored. u can use those for testing, or use the ones u wanna process (ofc, lol).

```
uv run scripts/transcribe_batch.py \
  --audio-dir /pl/active/blast-data/juan/outputs_podcasts/downloads/candace \
  --output-dir outputs/transcripts/candace \
  --language en
```

this is what i'm seeing after running the previous command:



```
rc computing ssh, juva3822
/scratch/alpine/juva3822/pypodcast main*
> uv run scripts/transcribe_batch.py \
> --audio-dir /pl/active/blast-data/juan/outputs_podcasts/downloads/candace \
> --output-dir outputs/transcripts/candace \
> --language en

Podcast Transcription Pipeline

discovering audio files...
found 320 audio files
skipping 7 files with existing transcripts
will process 313 new files

model: large-v3, compute: float16, device: cuda

System Information
  • CPU: 64 cores (64 threads)
  • Memory: 251.0 GB
  • GPU 0: NVIDIA H100 PCIe MIG 3g.40gb (40 GB)

[02/11/26 13:39:50] INFO    using gpu: NVIDIA H100 PCIe MIG 3g.40gb (39.5GB)
                    INFO    batch size: 31
                    INFO    loading whisperx model: large-v3

Resources
CPU      8.6% (64 cores)
[02/11/26 13:41:46] DEBUG   Registered checkpoint save hook for _speechbrain_save
                    DEBUG   Registered checkpoint load hook for _speechbrain_load
                    DEBUG   Registered checkpoint save hook for save
                    DEBUG   Registered checkpoint load hook for load
[02/11/26 13:41:49] INFO    Applied quirks (see `speechbrain.utils.quirks`): [disable_jit_profiling, allow_tf32]
                    INFO    Excluded quirks specified by the `SB_DISABLE_QUIRKS` environment (comma-separated list): []
                    DEBUG   Registered checkpoint save hook for _save
                    DEBUG   Registered checkpoint load hook for _recover
[02/11/26 13:41:51] INFO    HTTP Request: GET https://huggingface.co/api/models/Systran/faster-whisper-large-v3/revision/main
                    INFO    "HTTP/1.1 200 OK"

>>Performing voice activity detection using Pyannote...
[02/11/26 13:41:56] INFO    Lightning automatically upgraded your loaded checkpoint from v1.5.4 to v2.6.1. To apply the upgrade to
your files permanently, run `python -m pytorch_lightning.utilities.upgrade_checkpoint
.venv/lib/python3.12/site-packages/whisperx/assets/pytorch_model.bin`
Model was trained with pyannote.audio 0.0.1, yours is 3.4.0. Bad things might happen unless you revert pyannote.audio to 0.x.
Model was trained with torch 1.10.0+cu102, yours is 2.5.1+cu121. Bad things might happen unless you revert torch to 1.x.
                    INFO    model loaded in 125.9s
                    INFO    adjusted batch size: 31 -> 29
                    INFO    transcribing: 11_never_forget_but_also_never_remember_candace_ep_63.mp3
[02/11/26 13:43:25] INFO    transcribed 11_never_forget_but_also_never_remember_candace_ep_63.mp3 in 88.6s
                    INFO    saved transcript to
                    INFO    outputs/transcripts/candace/candace/11_never_forget_but_also_never_remember_candace_ep_63.json
                    INFO    adjusted batch size: 29 -> 26
                    INFO    transcribing: 1_podcast_worldwide_plus_erika_kirk_s_interview_candace_ep_260.mp3

Resources
CPU      36.0% (64 cores)
Memory   37.9%
          (95.1GB/251.0GB)
GPU      33.6%
          (13.3GB/39.5GB)

# transcribing | 1 ✓ 0 ✕ | 1/313 0% 0:03:53 -!-:-
[0] 0: bash$ "login-c14" 13:43 11-Feb-24
```

note

if the gpu utilization shows 0.0% or very low usage, that means that `pynvml` is likely not returning utilization for mig instances.

mig partitions often report 0% via `nvmldevicegetutilizationrates` because `nvml` doesn't support per-mig-instance utilization tracking. this is a known nvidia limitation 😞.

g) diarization

this is a sample command i used for my pipeline. i leave my hf token here, so u can use it if u want to. u should adapt the number of `workers` depending on how many cpus u asked for when running `sinteractive ... --ntasks=5`.

the `--compile-mode` is just an optimization for running things on (in ?) a gpu.

```
./scripts/diarize_all_podcasts.sh \  
  --transcripts-root outputs/transcripts/candace \  
  --audio-base-dir /pl/active/blast-data/juan/outputs_podcasts/downloads \  
  --output-dir outputs/diarizations/candace \  
  --hf-token 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 \  
  --workers 5 \  
  --compile-mode max-autotune
```

output:

rc computing ssh, juva3822

```
./scripts/diarize_all_podcasts.sh \
--transcripts-root outputs/transcripts/candace \
--audio-base-dir /p/active/blast-data/juan/outputs_podcasts/downloads \
--output-dir outputs/diarizations/candace \
--hf-token hf_VtWv1Z10YtGSDWfUobFqRHMtWYtnzpm \
--compile-mode max-autotune
```

Discovering podcasts in: outputs/transcripts/candace

Debug: Checking directory contents...

• json files in directory: 0
• Subdirectories found: 1

Multi-podcast mode: Looking for subdirectories with JSON files

Found 1 podcast subdirectories:
• candace (1 transcripts)

Processing all podcasts from root directory...
(This maintains correct relative paths for audio file resolution)

Podcast Diarization Pipeline

using CUDA: NVIDIA M100 PCIe MIG 3g.40gb (\$9.60B)

discovering files...

[02/11/26 14:22:10] INFO discovered 1 transcript-audio pairs

found 1 transcript-audio pairs

checking for existing RTTM files...

will process 1 new files

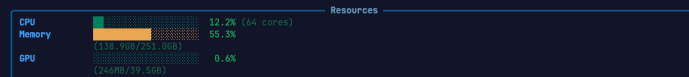
grouping files by duration...

created 1 batches of similar-duration files

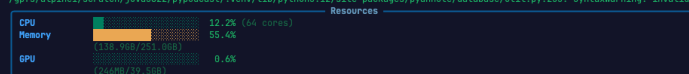
System Information

• CPU: 64 cores (64 threads)
• Memory: 251.0 GB
• GPU 0: NVIDIA M100 PCIe MIG 3g.40gb (40 GB)

processing files with 4 parallel workers...



diarization | starting... 0/1 0% 0:01:49 -::--//gpfs/alpine1/scratch/juva3822/pypodcast/.venv/lib/python3.12/site-packages/pyannote/database/util.py:182: SyntaxWarning: invalid escape sequence '\s'
sep="\s+",
/gpfs/alpine1/scratch/juva3822/pypodcast/.venv/lib/python3.12/site-packages/pyannote/database/util.py:216: SyntaxWarning: invalid escape sequence '\s'
sep="\s+",
/gpfs/alpine1/scratch/juva3822/pypodcast/.venv/lib/python3.12/site-packages/pyannote/database/util.py:253: SyntaxWarning: invalid escape sequence '\s'



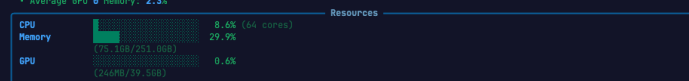
diarization | starting... 0/1 0% 0:02:00 -::--//gpfs/alpine1/scratch/juva3822/pypodcast/.venv/lib/python3.12/site-packages/pyannote/database/loader.py:93: SyntaxWarning: invalid escape sequence '\s'



✓ candace/10_candace_ep_248

Resource Usage Summary:

- Average CPU: 17.6%
- Average Memory: 56.0%
- Average GPU 0 Utilization: 0.0%
- Average GPU 0 Memory: 2.3%



diarization | 1 ✓ 0 ✖ 1/1 100% 0:03:49 0:00:00

✓ processing complete!

skipped (already exists): 0
total processed: 1
successful: 1
failed: 0

FINAL SUMMARY

Total podcasts discovered: 1
Total time: 0h 4m 26s

✓ Processing complete!

/scratch/alpine/juva3822/pypodcast main*
/scratch/alpine/juva3822/pypodcast main*

4m 26s juva3822@bpgu-g4-u10 02:26:03 PM

00 0:00:00

4m 26s juva3822@bpgu-g4-u10 02:26:03 PM